

Don't Shit where You Drill!



Don't Shit where You Drill!

Michel Hardenberg

PhD student at AU - Quantifying the Vadose Zone
Department of Electrical and Computer Engineering

**MELLEMFOLKELIGT
SAMVIRKE** **act:onaid**

CLIMATE JUSTICE SOCIETY SEPTEMBER 19, 2026

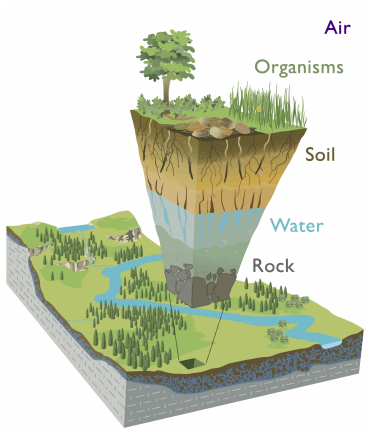
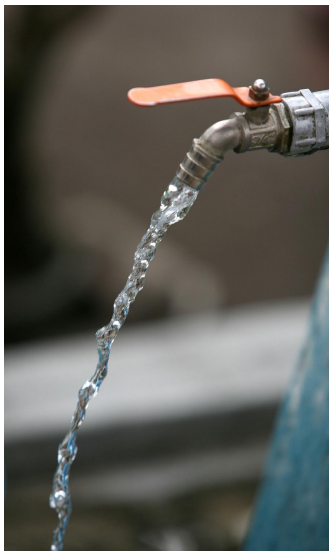


Illustration of soil layers in the near surface [Chorover et al., 2007].

ToC

Part One:

- ▶ What are the culprits?
- ▶ How are we looking?
- ▶ What is the government doing?
Promising?



GEUS WATER REPORT 2025



¹PFOS, PFOA, PFBS, PFHxS

GEUS WATER REPORT 2025

Pesticides are found in 70% of testcites. 1 out of 3 exceeds the safety limits.



¹PFOS, PFOA, PFBS, PFHxS

GEUS WATER REPORT 2025

Pesticides are found in 70% of testcites. 1 out of 3 exceeds the safety limits.

Nitrate from excessive farming reaches high concentrations in 1 out of 3 sites. 16% break the limits. In shallow depth sites 31% do.



¹PFOS, PFOA, PFBS, PFHxS

GEUS WATER REPORT 2025

Pesticides are found in 70% of testcites. 1 out of 3 exceeds the safety limits.

Nitrate from excessive farming reaches high concentrations in 1 out of 3 sites. 16% break the limits. In shallow depth sites 31% do.

PFAS can be measured in every third site. The four highly regulated compounds¹ in 24%.

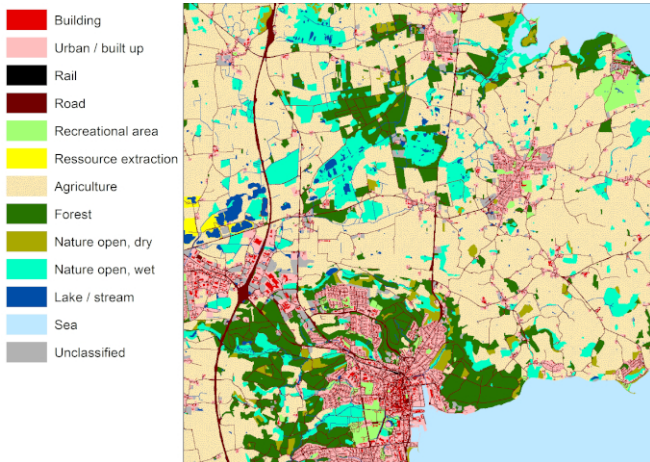


¹PFOS, PFOA, PFBS, PFHxS



Nitrate - the Classic





More than half of the area of Denmark is used for agriculture!²

²<https://www.dst.dk/en/Statistik/emner/miljoe-og-energi/areal/arealopgoerelser>

More than half of the area of Denmark is used for agriculture!³
Of that only 4 % for direct human consumption⁴!

³<https://www.dst.dk/en/Statistik/emner/miljoe-og-energi/areal/arealopgoerelser>

⁴<https://www.dyrenesbeskyttelse.dk/artikler/danmark-har-verdens-stoerste-koedindustri>

BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!



Foto: TV2 ØJ

Landmænd raser over nyt lovforslag om kvælstofudledning

**MELLEMFOLKELIGT
SAMVIRKE** **actionaid**

BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!



BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!

Food waste in Denmark

Sumtotal 881.062 tons / year

- Food industry 457,172 metric tons / year (51,9%)

Households 235,000 metric tons / year (26,7%)

Detail og engros 103,906 metric tons / year
(11,8%)

Primary production 42,984 metric tons / year (4,9
%)

Service sector 42,000 metric tons / year (4,8 %)

<https://stopspildafmad.org/om-madspild/madspild-i-tal/>

MEET THE MEAT THAT REALLY SHOULD MEET IT END



[https://www.dyrenesbeskyttelse.dk/artikler/
danmark-har-verdens-stoerste-koedindustri](https://www.dyrenesbeskyttelse.dk/artikler/danmark-har-verdens-stoerste-koedindustri)

MEET THE MEAT THAT REALLY SHOULD MEET IT END



or ~ 1.92 mio metric tons /
year. Exports $\sim 85\%$ to 90% ⁵

⁵[https://www.dyrenesbeskyttelse.dk/artikler/
danmark-har-verdens-stoerste-koedindustri](https://www.dyrenesbeskyttelse.dk/artikler/danmark-har-verdens-stoerste-koedindustri)

BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!



TV2

ØJ

- universitetets veje.

CONSEQUENCES

Environmental Pollution of bodies of water. Excess algae growth, extinction of fauna etc.

CONSEQUENCES FROM THE TAB

“Nitrate in drinking water is a contaminant which can affect human health and has been associated with an increased risk of, amongst other diseases, colorectal cancer”. [Jacobsen et al., 2024]

For over 60 years, the global guideline for nitrates in drinking water has been 50 milligrams per litre (mg/L) [...] set in the 1950s⁶.

⁶<https://www.greenpeace.org/international/story/82511/industrial-agriculture-profits-toxic-nitrates-pollution-tap-water/>

This limit is hopelessly out of date

Literature proposes lower limits of $< 10\%$ reducing the risk by 15%. Even lower limits of $< 4\%$ bring this number down further [Jacobsen et al., 2024, Schullehner et al., 2018].

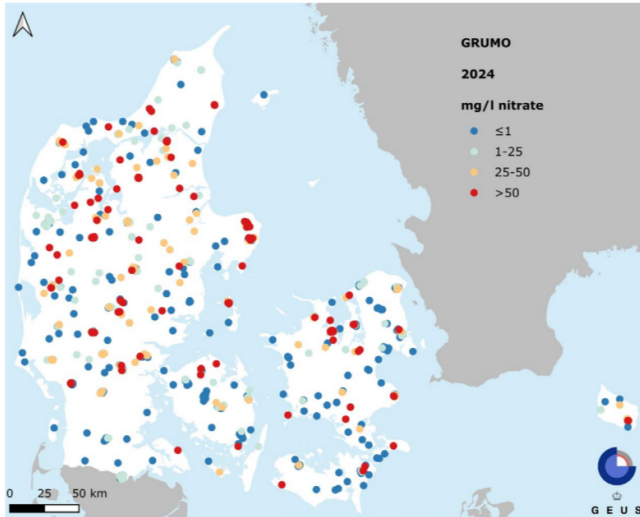
BLUE BABIES (NOT GOOD)



Infant methemoglobinemia

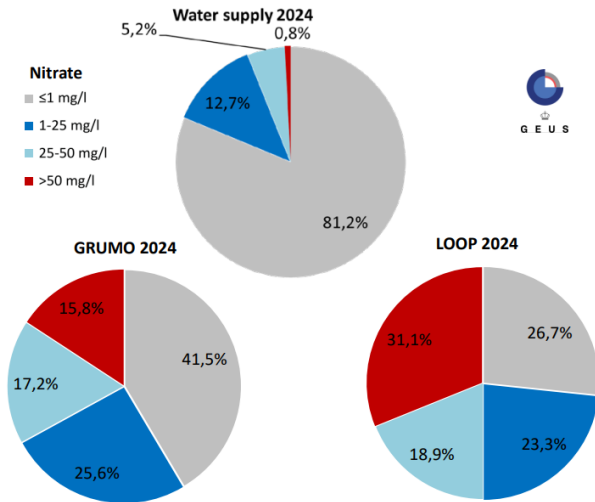
- ▶ Nitrate prevents oxygen from binding to their blood. Can progress to cause coma and death if not left untreated [Knobloch et al., 2000, Chen and Nappe, 2026].
- ▶ 2 cases where water for formula contained concentrations of 22.9 and 27.4 mg/L [Knobloch et al., 2000].

NITRATE POLLUTION IN DENMARK



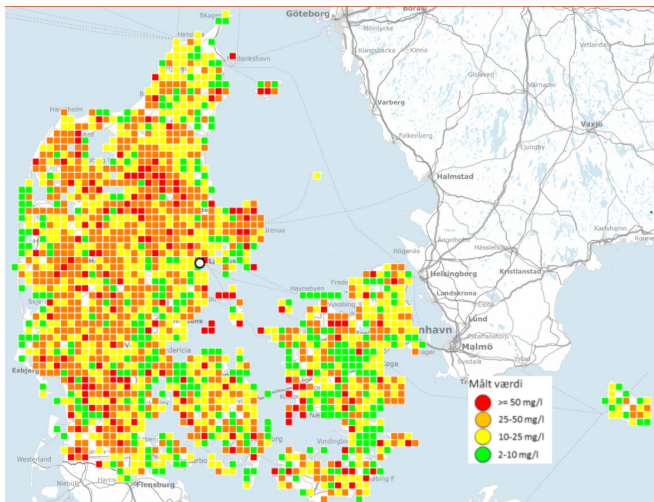
From GEUS' groundwater report 2025 [Lærke Thorling et al., 2025].

MIXING AND EXPLOITING



From GEUS' groundwater report 2025 [Lærke Thorling et al., 2025].

NITRATE POLLUTION IN DENMARK



<https://www.geus.dk/produkter-ydelser-og-faciliteter/data-og-kort/grundvandskort-og-data>



PFAS - the Immortal





WHERE DO THEY LIVE?

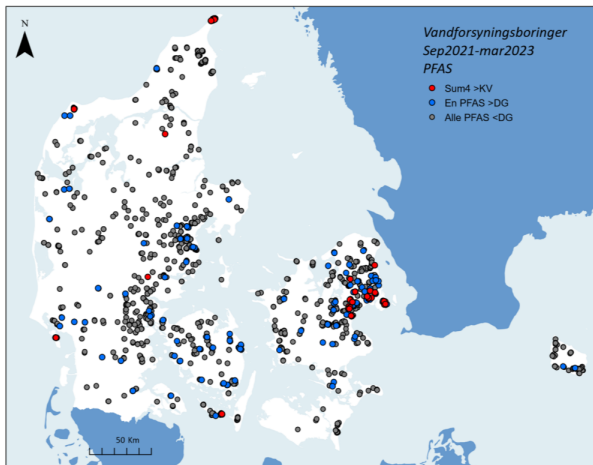
Where **DON'T** they live!? PFAS are found in water, air, fish, and soil..

WHERE DO THEY LIVE?

Where DON'T they live!? PFAS are found in water, air, fish, and soil..

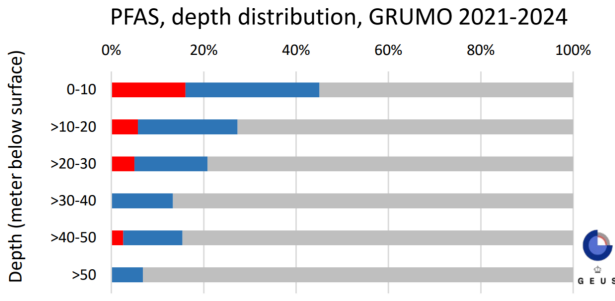
Also, in blood of people, animals, in food products⁷

PFAS MEASUREMENTS IN GROUNDWATER WELLS



Red: PFAS chemicals above limit. *Blue:* Measured PFAS chemicals. *Grey:* None detected [Anders R. Johnsen, Lærke Thorling & Christian N. Albers, 2023].

PFAS MEASUREMENTS IN GROUNDWATER WELLS



Red: PFAS chemicals above limit. *Blue:* Measured PFAS chemicals. *Grey:* None detected [Lærke Thorling et al., 2025].

PFAS does not simply break down. They're the eternity chemicals.

BIG PROMISES

- ▶ Protect 160 thousand hectar against pesti- and herbicides.
- ▶ 30 % of Denmark will be protected.
- ▶ Reduce environmental impact of especially pig farms.

BIG PROMISES

Already theres been strong push back by the Landbrug &
Fødevarer *[noa, 2026]*

BIG PROMISES

Already there's been strong push back by the Landbrug & Fødevarer *[noa, 2026]*

Environmental organisations like DN are cautiously optimistic *[noa, a]*

BIG PROMISES

Already there's been strong push back by the Landbrug & Fødevarer *[noa, 2026]*

Environmental organisations like DN are cautiously optimistic *[noa, a]*

The report is quite unspecific about e.g. PFAS and reduction of stock animals.

SPRØJTEFORBUD 2026

“Vi har knoklet for at blive klar til forhandlingerne om et nationalt sprøjteforbud. Det er nu, vi skal tage fat på at beskytte vores grundvand, så fremtidige generationer også kan drikke rent vand fra hanen. Målet er at blive enige om en aftale, så vi kan fremsætte et lovforslag senere på året. Beskyttelse af drikkevandet er min absolutte hovedprioritet og den første store opgave, jeg har kastet mig over som miljøminister.”

Maria Reumert Gjerding
Miljøminister

SPRØJTEFORBUD 2026



ToC

Part Two:

- ▶ Case studies in the global south
- ▶ How are we as Europeans involved?
- ▶ How is climate change changing the picture?

WATER IN KENYA



WATER IN KENYA

Article 42 of the Constitution of Kenya, 2010: “Every person has the right to a clean and healthy environment”, while Article 43 (d) of the Constitution guarantees the: “Right to clean and safe water in adequate quantities to every person.”

SOME NUMBERS [noa, b]

- ▶ Only about 15 percent of Kenya's water resources are easily accessible, and the majority of that is surface water.

SOME NUMBERS [noa, b]

- ▶ Only about 15 percent of Kenya's water resources are easily accessible, and the majority of that is surface water.
- ▶ Total abstraction 7 million m³/year - a thenth of what is available. Recharge about 8% to 9% of total.

SOME NUMBERS [noa, b]

- ▶ Only about 15 percent of Kenya's water resources are easily accessible, and the majority of that is surface water.
- ▶ Total abstraction 7 million m³/year - a thenth of what is available. Recharge about 8% to 9% of total.
- ▶ still aquifers are over abstracted - water level decline and quality deterioration.

SOME NUMBERS [noa, b]

- ▶ Only about 15 percent of Kenya's water resources are easily accessible, and the majority of that is surface water.
- ▶ Total abstraction 7 million m³/year - a thenth of what is available. Recharge about 8% to 9% of total.
- ▶ still aquifers are over abstracted - water level decline and quality deterioration.
- ▶ Poor legislation and mapping - 35.000 unregistered boreholes

WATER IN KENYA



Black water river running through the Kenyan slum next to Dandora landfill
©Christina Anderskov

WATER IN KENYA

Kenya generates between 3,000 to 4,000 tons of waste per day.

Nairobi generates between 2,000 to 2,500 tons of waste daily.

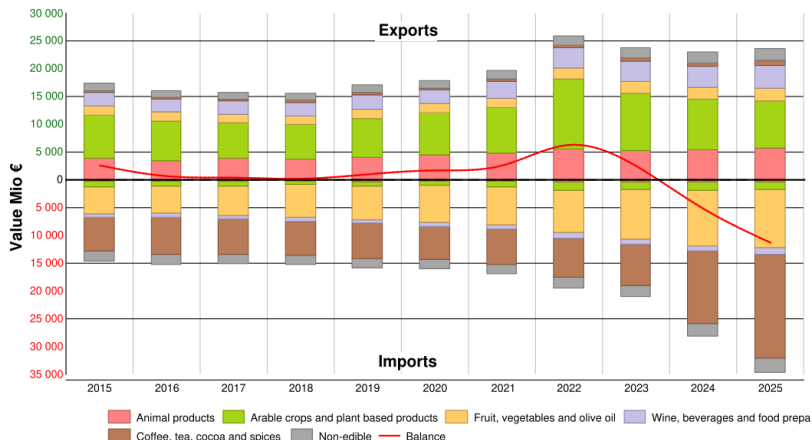
Less than half of that amount is re- or upcycled.

Dumped in: landfills, along roads, in streams or rivers or burned.

This happens with little awareness of the subsurface and thus groundwater.⁸

⁸<https://eng.mst.dk/about-the-danish-epa/global-cooperation/kenyacircular-economy>

TRADE WITH AFRICA



Import from and exports to Africa⁹

⁹Data from https://agriculture.ec.europa.eu/system/files/2023-05/agrifood-africa-all-countries_en.pdf

BANGLADESH

‘I fear people will go to war over water’: as wells run dry, farmers struggle to survive in Bangladesh// - the Guardian 2026

BANGLADESH



Groundwater use¹⁰:

- ▶ 90% for agriculture
- ▶ 8–9% in homes
- ▶ 1–2% used industrial especially fast fashion

¹⁰ <https://un-igrac.org/data/country-profiles/bangladesh/>

- ▶ Massive pollution and public safety from e-waste handling *[Faisal et al., 2025]*

¹¹<https://esdo.org/wp-content/uploads/2023/03/A-Country-Situation-Report-on-Transboundary-Movement-of-Plastic-Waste.pdf>

- ▶ Massive pollution and public safety from e-waste handling *[Faisal et al., 2025]*
- ▶ Tremendous (illegal) imports of plastics and metal waste¹¹

¹¹<https://esdo.org/wp-content/uploads/2023/03/A-Country-Situation-Report-on-Transboundary-Movement-of-Plastic-Waste.pdf>

Climate - It's coming for you



Flood move a great amount of pollutants into water resources and move natural barriers.

2025 GREECE

Kort nyt | 3. jul 2025

Græsk ferieø hærget af skovbrand - igen



Omkring 3.000 beboere og turister er blevet evakueret. (Foto: © STEFANOS RAPANIS, Ritzau Scanpix)

Fires require access to great amounts of water for controlling and preventing them.

CONGO 2026

Jordskred ved mine i DR Congo dræber over 200 mennesker - mange af dem børn



Billedet her af Rubaya-minen i det østlige DR Congo er fra den 30. januar, hvor minen ligeledes blev ramt af et jordskred. Minen kontrolleres af oprørsgruppen M23 og står for en betydelig del af verdens produktion af coltan, der bruges i elektronik som mobiltelefoner og computere. (Foto: © AFP, Ritzau Scanpix)

Cause destruction and pollution of water sources and infrastructure

Particularly affects vulnerable populations in the Global South *[Ngcamu, 2023]*

Particularly affects vulnerable populations in the Global South *[Ngcamu, 2023]*

- ▶ Increases in droughts and extremes wheather
- ▶ Unable to protect themselves
- ▶ systematicly overlooked

REFERENCES I



Eksisterer grøn trepart stadig? DN er ikke i tvivl.



Kenya water.



(2026).

Grøn trepart kan være i fare, advarer Landbrug & Fødevarer.

Section: Valg 2026.



Anders R. Johnsen, Lærke Thorling & Christian N. Albers (2023).

PFAS-stoffer og pesticider i grundvand Nationalt overblik 2023 til brug for InSa Drikkevand.



Chen, R. J. and Nappe, T. M. (2026).

Methemoglobinemia.

In *StatPearls*. StatPearls Publishing, Treasure Island (FL).



Chorover, J., Kretzschmar, R., Garcia-Pichel, F., and Sparks, D. L. (2007).

Soil Biogeochemical Processes within the Critical Zone.

Elements, 3(5):321–326.



Faisal, A. K. M., Ankhi, I. J., Hossain, G. A., Ahmed, M. M., Siddhpura, M., and Masud, M. H. (2025).

Emerging concerns associated with E-waste exposure in Bangladesh.

Environmental Science and Pollution Research, 32(14):8595–8628.

REFERENCES II



Jacobsen, B. H., Hansen, B., and Schullehner, J. (2024).

Health-economic valuation of lowering nitrate standards in drinking water related to colorectal cancer in Denmark.

Science of The Total Environment, 906:167368.



Knobeloch, L., Salna, B., Hogan, A., Postle, J., and Anderson, H. (2000).

Blue babies and nitrate-contaminated well water.

Environmental Health Perspectives, 108(7):675–678.



Lærke Thorling, Christian Nyrop Albers, Birgitte Hansen, Anders R. Johnsen, Jacob Kidmose, Mette Hilleke Mortensen, Lars Troldborg, Denitza Voutchkova, and Lærke Thorling (2025).

Groundwater monitoring, Status and trends 1989–2024, Summary.



Ngcamu, B. S. (2023).

Climate change effects on vulnerable populations in the Global South: a systematic review.

Natural Hazards, 118(2):977–991.



Schullehner, J., Hansen, B., Thygesen, M., Pedersen, C. B., and Sigsgaard, T. (2018).

Nitrate in drinking water and colorectal cancer risk: A nationwide population-based cohort study.

International Journal of Cancer, 143(1):73–79.